

About Mobiloitte

Mobiloitte's strength lies in its ability to know where IoT applications are needed as well as in finding faster and cheaper ways to connect with devices. The company also focuses on greener manufacturing and developing solutions with completely remote mobile device management (MDM), thus reducing device management complexity. Instead of just delivering cheaper solutions, Delhi-based Mobiloitte helps enterprises make affordable IoT solutions without compromising on security and interoperability. It develops products for predictive prevention of health issues, and to improve machine efficiency and reliability.

Q How is IoT enhancing the healthcare sector in India currently?

The Indian government is increasing its spending on healthcare, and closing the infrastructure gap as well as addressing the workforce scarcity. But to bring IoT to the healthcare sector, many big organisations have just started contributing with end-to-end patient engagement platforms. Wearables are also leading to innovations that enhance the fitness of a wider proportion of the population.

Apart from some large companies, startups are also actively researching and partnering up with NGOs to improve the health of the masses, using IoT.

Since 2012, the Indian health sector has witnessed active participation by IoT developers, resulting in solutions that address the problems arising from the 21st century's stress driven lifestyle. Wrist bands that add a dash of wearable fun to the fitness process are a huge hit among the health-conscious youth. Sensor-powered pendants that can detect falls are also available to support elderly people.

Q What are the big challenges in developing apps for the IoT world?

Just like building solutions in any other nascent space, creating apps for IoT also involves some major challenges.

Developers need to focus on network latency, determinism and bandwidth when building IoT apps for precision-based machines. If the timing is off, even for a millisecond, the entire production floor could fail. Likewise, IoT systems need to be adaptive and scalable through software. Having black box systems that do not communicate well together are of no use in the connected space.

IoT applications can be built on tens of thousands of sensor nodes, but all this increases threat surface areas by orders of magnitude. Therefore, developers need to implement security at each and every level of their code. Industrial systems also have to be continually

modified and maintained to meet changing requirements. Additionally, an effective platform-based approach should focus not on hardware or software but on the innovation within the application itself, and provide enough flexibility to the system to evolve and adapt.

Q How does your team resolve challenges in building IoT apps?

A single silver bullet cannot meet all the challenges in the IoT space. However, we consider that the challenges in developing apps for the connected space are forecastable —their redundancy is built-in and alliances with trusted and highly skilled organisations should be formed. We also prefer the establishment and formalisation of checklists and globally acceptable protocols in order to align different tools, technologies, people and processes.

Q Do you leverage any open source technologies while developing apps for IoT hardware?

The AllSeen Alliance, DSA, Eclipse IoT (Kura), and Open Connectivity Foundation are some of the promising open source technologies and platforms for connected devices. However, no single technology fits all solutions.

Hence, we believe in creating a hybrid architecture, which is the consolidation of the best of the breed from a plethora of open source technologies, as a one-stop solution for most IoT needs.

Q Is it easy to create a network of IoT devices through open source?

This would vary from case to case. While open source solutions can address some challenges, proprietary solutions can address other use cases, and several other use cases can be addressed via a mix of open source and proprietary solutions. As a thumb rule, the best-of-breed solutions need to be leveraged to create a seamless IoT platform that is universally acceptable, works in real-time, is cost effective and predictive.

Q Can a simple app developer be an enabler in the IoT world?

Yes, why not? The transformation of a traditional app developer to a developer for the IoT world requires only a few steps. We make this possible at Mobiloitte and create a pool of developers by training them, monitoring their progress on a weekly basis, and filling in their skill gaps by stepping in and providing thought leadership.

We have our own CoE (Centre of Excellence) managed by the Training Resources Group (TRG) and led by our CTO that continues to conduct research on new and upcoming technologies, creates reusable components, and develops zero build reusable frameworks as well as ready-to-use code snippets and